



Diesel Generator Set

mtu 20V4000 DS2750

380V – 11 kV/50 Hz/data center continuous power/
NOx emission optimized/20V4000G14F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits

- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support

- Global product support offered

Standards

- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating

- System ratings: 2530 kVA - 2660 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)

- Engine-generator set tested to ISO 8528-5 for transient response
- 100% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available

- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions

- NOx emission optimized

Certifications

- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary



A Rolls-Royce
solution

Application data ¹⁾

Engine			Liquid capacity (lubrication)	
Manufacturer		<i>mtu</i>	Total oil system capacity: l	390
Model		20V4000G14F	Engine jacket water capacity: l	205
Type		4-cycle	Intercooler coolant capacity: l	50
Arrangement		20V	Combustion air requirements	
Displacement: l		95.4	Combustion air volume: m³/s	3.2
Bore: mm		170	Max. air intake restriction: mbar	50
Stroke: mm		210	Cooling/radiator system	
Compression ratio		16.4	Coolant flow rate (HT circuit): m³/hr	80
Rated speed: rpm		1500	Coolant flow rate (LT circuit): m³/hr	32.5
Engine governor		ECU 9	Heat rejection to coolant: kW	970
Max power: kWm		2200	Heat radiated to charge air cooling: kW	470
Air cleaner		dry	Heat radiated to ambient: kW	105
Fuel system			Fan power for electr. radiator (40°C): kW	70
Maximum fuel lift: m		5	Exhaust system	
Total fuel flow: l/min		27	Exhaust gas temp. (after turbocharger): °C	530
Fuel consumption ²⁾			Exhaust gas volume: m³/s	8.7
At 100% of power rating:	l/hr	g/kwh	Maximum allowable back pressure: mbar	85
At 75% of power rating:	585.8	221	Minimum allowable back pressure: mbar	30
At 50% of power rating:	423.4	213		
	284.9	215		

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	NOx emission optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA52.3 UL16 (Low voltage Leroy Somer standard)	380 V	2112	2640	4011	2048	2560	3890
	400 V	2112	2640	3811	2048	2560	3695
	415 V	2112	2640	3673	2048	2560	3561
Marathon 1020FDL7093 (Low voltage Marathon)	380 V	2128	2660	4041	2064	2580	3920
	400 V	2096	2620	3782	2024	2530	3652
	415 V	2096	2620	3645	2024	2530	3520
Marathon 1030FDL7094 (Low voltage Marathon oversized)	380 V	2128	2660	4041	2064	2580	3920
	400 V	2096	2620	3782	2024	2530	3652
	415 V	2096	2620	3645	2024	2530	3520
Marathon 1030FDH7101 (Medium volt. marathon)	11 kV	2112	2640	139	2040	2550	134
Leroy Somer LSA53.2 ZL12 (Med. volt. Leroy Somer)	11 kV	2112	2640	139	2048	2560	134

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).
2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.